# Exit Ticket · reflect and retrieve

## Exit Ticket

Using low=mid+1 after a[mid]<target can prevent non-shrinking binary-search intervals.

## Use key vocabulary from memory

1. Explain off-by-one error in your own words.

2. Explain exception in your own words.

3. Explain trace in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 210 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Debugging and error correction

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Write −1 in 8-bit two's complement.

### Q2

What is the last index in a four-item array?

### Q3

What precondition does binary search need?

### Q4

What happens if code reads a[3] from a three-element C# array?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Check valid indexes — A three-element array has valid indexes 0, 1 and 2.

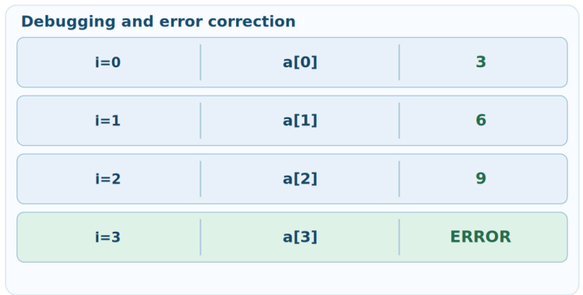
## Core idea 2

Follow the condition — When i is 3, i<=a.Length remains true and attempts to read a[3], causing an exception.

## Core idea 3

Fix and retest — Use i<a.Length, which stops before i becomes an invalid index.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Choose normal, boundary and erroneous indexes when a method accepts only indexes 0–4.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Fix the loop condition i<=a.Length when visiting every array item.

## Question 2

When a[mid]>target, what should binary search do to high?

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

off-by-one error — Loop repeats once too many or few times.

exception — Runtime failure or signal raised by an invalid operation.

trace — Record changing execution state.

precondition — Input requirement an algorithm assumes.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Correct a loop summing [2,4] that uses <= length.

## Follow the worked method

1. Predict — The array's valid indexes are 0 and 1, not 2.
2. Explain — Use i < values.Length and the total is 6.

**Source model · final illustrated step**

**Correct a loop summing [2,4] that uses <= length.**

| | | |
|---|---|---|
| i=0 | value 2 | total 2 |
| i=1 | value 4 | total 6 |
| i=2 | invalid | ERROR |

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict
With low=3 and high=4, midpoint remains 3 and low=mid fails to shrink the interval.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.